Boyce-Codd Normal Form

DB normal forms



- Domain/key
- 6th
- 5th = Project-join
- 4th
- Boyce-Codd = 3.5^{th}
- 3rd
- 2nd
- 1st

Based on functional dependencies, i.e., what attributes depend upon.

Goal: Prevent DB anomalies.

FD-based table normalization

While there are “bad” FDs

Decompose a table with “bad” FDs into sub-tables.

Boyce-Codd normal form (BCNF)

(BCNF) *"Each attribute must represent a fact about the key, the whole key, and nothing but the key."*

– Chris Date

Ignoring trivial FDs,

- Good: $X \to R$ X is a superkey
- Bad: $X \to A$ for $A \subset R$ X is not a superkey

BCNF – definition

Relation R is in BCNF

iff

for every FD $A \rightarrow B$ in R,

$A \rightarrow B$ is trivial, or

A is a superkey for R.

Relation R is in BCNF:
for every FD $A \rightarrow B$ in R,
$A \rightarrow B$ is trivial, or
A is a superkey for R.

Activity

name	ssn	phone	city
Mary	123-45-6789	713-555-1234	Houston
Mary	123-45-6789	713-555-6543	Houston
Joe	987-65-4321	512-555-2121	Austin
Joe	987-65-4321	512-555-1234	Austin

Find a FD that holds for this data, but that doesn't satisfy BCNF.

I'll pick {name} $\rightarrow$ {ssn, city}.

Fixing the example

Decompose table into sub-tables based on {name} → {ssn, city}.

name	ssn	city
Mary	123-45-6789	Houston
Joe	987-65-4321	Austin

name	phone
Mary	713-555-1234
Mary	713-555-6543
Joe	512-555-2121
Joe	512-555-1234

BCNF decomposition – algorithm

$BCNF_F(R)$ =

Find FD $A \rightarrow B$ where $A^+ \neq A$ (not trivial) and $A^+ \neq R$ (not superkey).

If no such A, then return R.

Let $D = A^+ - A$. (attributes functionally determined by A)

Let $N = R - A^+$. (attributes not functionally determined by A)

Decompose R into two tables: $R_1(A \cup D)$, $R_2(A \cup N)$.

Return $BCNF_F(R_1)$, $BCNF_F(R_2)$.

BCNF decomposition

$BCNF_F(R)$ =

Find FD $A \rightarrow B$ where $A^+ \neq A$ and $A^+ \neq R$.

If no such A, then return R.

Let $D = A^+ - A$.

Let $N = R - A^+$.

Decompose R into $R_1(A \cup D)$, $R_2(A \cup N)$.

Return $BCNF_F(R_1)$, $BCNF_F(R_2)$.

R(a,b,c,d,e)
{a} → {b, c}
{c} → {d}

A = {a}
A^+ = {a,b,c,d}

D = {b,c,d}
N = {e}

R_1(a,b,c,d)
R_2(a,e)

BCNF decomposition

$BCNF_F(R)$ =

Find FD $A \rightarrow B$ where $A^+ \neq A$ and $A^+ \neq R$.

If no such A, then return R.

Let $D = A^+ - A$.

Let $N = R - A^+$.

Decompose R into $R_1(A \cup D)$, $R_2(A \cup N)$.

Return $BCNF_F(R_1)$, $BCNF_F(R_2)$.

R_1(a,b,c,d)
{a} → {b, c}
{c} → {d}

A = {c}
A^+ = {c,d}

D = {d}
N = {a,b}

R_{11}(c,d)
R_{12}(a,b,c)

BCNF decomposition

R_{11}(c,d)
{a} → {b, c}
{c} → {d}

$BCNF_F(R)$ =

Find FD $A \rightarrow B$ where $A^+ \neq A$ and $A^+ \neq R$.

If no such A, then return R.

No such A.

Let $D = A^+ - A$.

Let $N = R - A^+$.

Decompose R into $R_1(A \cup D)$, $R_2(A \cup N)$.

Return $BCNF_F(R_1)$, $BCNF_F(R_2)$.

BCNF decomposition

R_{12}(a,b,c)
{a} → {b, c}
{c} → {d}

$BCNF_F(R)$ =

Find FD $A \rightarrow B$ where $A^+ \neq A$ and $A^+ \neq R$.

If no such A, then return R.

No such A.

Let $D = A^+ - A$.

Let $N = R - A^+$.

Decompose R into $R_1(A \cup D)$, $R_2(A \cup N)$.

Return $BCNF_F(R_1)$, $BCNF_F(R_2)$.

BCNF decomposition

$BCNF_F(R)$ =

Find FD $A \rightarrow B$ where $A^+ \neq A$ and $A^+ \neq R$.

If no such A, then return R.

Let $D = A^+ - A$.

Let $N = R - A^+$.

Decompose R into $R_1(A \cup D)$, $R_2(A \cup N)$.

Return $BCNF_F(R_1)$, $BCNF_F(R_2)$.

R_2(a,e)
{a} → {b, c}
{c} → {d}

No such A.

BCNF example summary

